



**College of Computing and Informatics
Department of Computer Science**

Course Syllabus

Course Title

Object Oriented Software Design & Implementation

Course Code
1501361

Spring 2025/2026

Course Syllabus

Course Title	Object Oriented Software Design & Implementation				
Course Code	1501361				
Credit Hours	3	Contact Hours	3		
Course Category	Program required				
Course Type	Compulsory				
Pre-requisite(s)	1501214 - Programming With Data Structures OR 1410214 - Programming With Data Structures				
Co-requisite(s)					
Semester	Spring 2025/2026				
Instructor Name	Email	Classroom	Section	Session Type	Time
Isam Mashhour Hasan Al Jawarneh	ijawarneh@sharjah.ac.ae	C14-W8-STU008M	61	Lecture Undergraduate	MW 1100 - 1215

Course Description:

The course teaches the essential skills in object-oriented analysis and design in Universal Modeling Language (UML). The course introduces the students to the software life cycle, requirements, analysis, system design, and implementation.

Course Objectives/Goals:

The goals of this Course are to:

- Understand the key points of Object Oriented analysis and design.
- Design, implement, and test an Object Oriented application.
- Use UML in designing software systems.
- Create an implementation based on Object Oriented design.

Course Learning Outcomes:

By the end of successful completion of this Course, the student will be able to:

1. Apply object oriented principles: Abstraction, Encapsulation, Modularity & Hierarchy.
2. Master object oriented analysis activities.
3. Apply structural and behavioral design concepts.
4. Use UML in software analysis and design.
5. Use design patterns in software design.
6. Develop an object-oriented prototype using specific object oriented programming language.

Alignment of Course Student Learning Outcomes to Program Student Learning Outcomes

Program Learning Outcomes (PLOs)	Course Learning Outcomes (CLOs)
Analyze a complex computing problem and apply principles of computing and other relevant	

Program Learning Outcomes (PLOs)	Course Learning Outcomes (CLOs)
disciplines to identify solutions.	- Apply object oriented principles: Abstraction, Encapsulation, Modularity & Hierarchy.
Design, implement, and evaluate a computing and multimedia-based solution to meet a given set of computing requirements in the context of the program's discipline.	- Use UML in software analysis and design.
Design, implement, and evaluate a computing and multimedia-based solution to meet a given set of computing requirements in the context of the program's discipline.	- Use design patterns in software design.
Design, implement, and evaluate a computing and multimedia-based solution to meet a given set of computing requirements in the context of the program's discipline.	- Develop an object-oriented prototype using specific object oriented programming language.
Identify and analyze user needs and to take them into account in the selection, creation, integration, evaluation, and administration of computing-based system.	- Master object oriented analysis activities.
Analyze a complex computing problem and apply principles of computing and other relevant disciplines to identify solutions.	- Apply structural and behavioral design concepts.

Weekly Distribution of Course Topics/Contents

Week	Topic	Comments*	CLOs
1	Fundamentals Of Object Oriented Concepts		1
2	- Abstract Data Types/Encapsulation/Information Hiding/Coupling/Cohesion - Topics on Inheritance/Polymorphism/Dynamic Binding		1
3	The Requirements Discipline		2
4	The Requirements Discipline		2,3,4
5	Object Oriented Analysis & Design, UML concepts		2,3,4
6	Object Oriented Analysis & Design, UML concepts		3,4
7	Object Oriented Analysis & Design, UML concepts		3,4
8	- Structural Modeling - Behavioral Modeling		4,5
9	- Structural Modeling - Behavioral Modeling		4,5
10	- Object Oriented Detailed Design - Basics of Object-Oriented Design Patterns		6
11	- Object Oriented Detailed Design - Basics of Object-Oriented Design Patterns		6
12	GUI Fundamentals & Applications GUI Tool Box (Forms, Labels, Textboxes, Buttons, List Box, Radio Buttons, Check boxes, Message Boxes...etc.,)		6
13	- Conditional Logic & Loops - Arrays - Classes & Objects		6
14	- Conditional Logic & Loops - Arrays		6

Week	Topic	Comments*	CLOs
	- Classes & Objects		
15	Object Oriented Programming: Project		6
16	Final Exam		1,2,3,4,5

Information on out-of-class assignments with due dates for submission

Assignment/Activity	Due Date	Comments
Project	By the end of 14th week	
Class Activity	Weeks 4, 9, 13	

Students' Assessment:

Students are assessed as follows:

Assessment Tool(s)**	Date	Weight (%)
Class Activity	Weeks 4, 9, 13	15
Projects	Week 14	20
Midterm Exam	Week 9	20
Final Exam	Week 16	45
Total		100

Course Outcome Assessment Plan:

Course LOs	Teaching/Learning Method(s)	Assessment Tool(s)	Performance Indicators	UoS competencies	Other competencies
Apply object oriented principles: Abstraction, Encapsulation, Modularity & Hierarchy.	- Problem-solving - Team discussion	- Exams - Projects - Class Activity	70% of student achieve 70% of the CLO	- Oral communication - Written communication - Teamwork or collaboration - Digital literacy	
Master object oriented analysis activities.	- Lectures - Problem-solving - Team discussion	- Exams - Projects - Class Activity	70% of student achieve 70% of the CLO	- Written communication - Teamwork or collaboration - Problem solving	
Apply structural and behavioral design concepts.	- Lectures - Problem-solving - Team discussion	- Exams - Projects - Class Activity	70% of student achieve 70% of the CLO	- Written communication	

Course LOs	Teaching/Learning Method(s)	Assessment Tool(s)	Performance Indicators	UoS competencies	Other competencies
				- Teamwork or collaboration - Problem solving	
Use UML in software analysis and design.	- Lectures - Problem-solving - Team discussion	- Exams - Projects - Class Activity	70% of student achieve 70% of the CLO	- Written communication - Teamwork or collaboration - Problem solving	
Use design patterns in software design.	- Lectures - Problem-solving - Team discussion	- Exams - Projects - Class Activity	70% of student achieve 70% of the CLO	- Written communication - Teamwork or collaboration - Problem solving	
Develop an object-oriented prototype using specific object oriented programming language.	- Problem-solving - Team discussion	- Projects	70% of student achieve 70% of the CLO	- Critical and analytical thinking - Teamwork or collaboration - Innovation / creative thinking - Problem solving	

Scope and Nature of Assessment Tools:

The aim of this section is to clarify what is required and how to submit, grade and assess each assessment tool:

Exams:

This assessment tool includes two exams, the midterm exam (20%), and the final exam (45%). The midterm exam covers the course's first 4 learning outcomes and includes material of the first 5 chapters. The final exam covers all course learning outcomes and includes all course material.

Class Activity:

There will be several class activities. Each one has an equal weight in the computation of the course grade. Class Activity assignments will be distributed on the eLearning system.

Project

Students work on a project in teams. A project includes Design and Implementation phases of a software cycle. The instructor gives the students a set of tools (see the Required Teaching Resources section) to be used in their project. At the end of the semester, the students submit a softcopy of their project and a report. Furthermore, the students must demonstrate and discuss their project in class or in the office. The projects are evaluated

based on the Design and Implementation phases of the software cycle. In addition, students are evaluated based on their report, demo, and contribution to the project.

Teaching and Learning Resources:

Textbook(s):

Dathan, B., & Ramnath, S.. (2015). Object-Oriented Analysis, Design and Implementation: An Integrated Approach. - ed. Springer.

Other Resources:

- UML tools used for UML design.
- Java.

Attendance policy:

Attendance is compulsory. A student missing 10% of the total allocated course hours will receive 1st warning notice and a student missing 15% will receive 2nd warning notice. A student missing 20% will be forced to withdraw (in accordance with the university regulations).

Plagiarism/Cheating:

Students are expected to do their own work. You are allowed to work on assignments in teams only if specified by the instructor. In other words, students are encouraged to communicate about general principles of the course, but all assigned homework must be done on an individual basis. The instructor is available to provide any assistance that you may need. Cheating is considered a serious offense by the university. You should be aware of the severe penalty for cheating (refer to the student code of conduct published in the university catalogue).

Grading rubrics:

Grading Rubric for <i>Course Projects</i>					
Scaled Score:		1/0	2	3	4
Criteria	Weight	Below Standard	Approaching Standard	At Standard	Exceeds Standard
Introduction	10%	No clear introduction, absent, or no evidence.	The introduction states the main topic but does not adequately preview the structure of the report.	The introduction states the main topic and previews the structure of the report.	The introduction is engaging, states the main topic and previews the structure of the report.
Data	20%	Data are inadequate and selected following the wrong procedure.	Data are selected but procedure of data selection is not followed.	Data selected following the standard procedure but not described in report.	Data are correctly selected following standard procedure and adequately described in report.
Analysis and Reporting of Results (if	20%	Calculated wrong measures and analysis not relevant.	Calculated correct measures but analyses are not correct.	Calculated correct measures but analyses are not	Calculated correct measures and analyses are correct and provide more

Grading Rubric for Course Projects					
Scaled Score:		1/0	2	3	4
Criteria	Weight	Below Standard	Approaching Standard	At Standard	Exceeds Standard
applicable)				adequate.	insights.
Authenticity and Citation*	20-30%	Some contents are paraphrased; plagiarism is most likely.	Some content is paraphrased; plagiarism is likely.	Some content is paraphrased and no suspicion of plagiarism.	All contents are original and well written.
Writing	20-30%	Errors in language that cause the content to be mostly incomprehensible.	Some errors in language that often inhibit comprehension.	Some errors in language that do not often inhibit comprehension.	Almost no error in language and well written.
Teamwork	10%	Team did not collaborate or communicate well.	Team did not collaborate or communicate well. A lack of mutual respect was frequently noted.	Team did collaborate or communicate well.	Team did collaborate or communicate well with high level mutual respect.

Grading Rubric for Course Projects					
Scaled Score:		1/0	2	3	4
Criteria	Weight	Below Standard	Approaching Standard	At Standard	Exceeds Standard
Introduction	10%	No clear introduction, absent, or no evidence.	The introduction states the main topic but does not adequately preview the structure of the report.	The introduction states the main topic and previews the structure of the report.	The introduction is engaging, states the main topic and previews the structure of the report.
Data	20%	Data are inadequate and selected following wrong procedure.	Data are selected but procedure of data selection is not followed.	Data selected following standard procedure but not described in report.	Data are correctly selected following standard procedure and adequately described in report.
Analysis and Reporting of Results (if applicable)	20%	Calculated wrong measures and analysis not relevant.	Calculated correct measures but analyses are not correct.	Calculated correct measures but analyses are not adequate.	Calculated correct measures and analyses are correct and provide more insights.
Authenticity and Citation*	20-30%	Some contents are paraphrased; plagiarism is most likely.	Some content is paraphrased; plagiarism is likely.	Some content is paraphrased and no suspicion of plagiarism.	All contents are original and well written.
Writing	20-30%	Errors in language that cause the content to be mostly incomprehensible.	Some errors in language that often inhibit comprehension.	Some errors in language that do not often inhibit comprehension.	Almost no error in language and well written.

Grading Rubric for <i>Course Projects</i>					
Scaled Score:		1/0	2	3	4
Criteria	Weight	Below Standard	Approaching Standard	At Standard	Exceeds Standard
Team work	10%	Team did not collaborate or communicate well.	Team did not collaborate or communicate well. A lack of mutual respect was frequently noted.	Team did collaborate or communicate well.	Team did collaborate or communicate well with high level mutual respect.